# The Paris Targets as Numbers

> Part of the Functional Series. The Agreement states its goals as temperatures. To compare them with emissions data, they have to be turned into quantities: how much can still be emitted, how fast emissions must fall, and by when. This document does that translation and lays the platform's pathways next to it. Where the platform's own numbers are used, they come from the scenarios described in Scenarios and Temperature Mapping.

## 1. The goals in numbers

| Goal | In numbers | Source |
|---|---|---|
| Temperature | Well below 2 °C, pursue 1.5 °C, above 1850–1900 | Paris Agreement, Article 2.1(a) |
| Where warming stands | About 1.4 °C long-term (observed 2023–2025 average about 1.5 °C) | Copernicus; Forster et al. 2026 (human-induced 1.37 °C in 2025) |
| Emissions in 2030 | GHG –43% (range –34 to –60%) and $CO_2$ –48% from 2019 for 1.5 °C; GHG –21% and $CO_2$ –22% for 2 °C | IPCC AR6 Synthesis Report, Table SPM.1 |
| Emissions in 2035 | GHG –60% and $CO_2$ –65% (1.5 °C); GHG –35% and $CO_2$ –37% (2 °C) | IPCC AR6 Synthesis Report, Table SPM.1 |
| 2035 in UNEP's pathway set | –55% (1.5 °C) and –35% (2 °C) in total GHG | UNEP *Emissions Gap Report 2025* |
| Net zero $CO_2$ | Early 2050s (1.5 °C); early 2070s (2 °C) | IPCC AR6 Synthesis Report B.6.1 |
| Net zero GHG | Around the 2070s in 1.5 °C pathways that reach it | IPCC AR6 Synthesis Report B.6.1 |
| Remaining carbon budget, 1.5 °C (50%) | 130 $GtCO_2$ (Forster et al.) or 170 $GtCO_2$ (Global Carbon Budget), from the start of 2026 | See Benchmarks |
| Remaining carbon budget, 2 °C (50%) | about 1,050 $GtCO_2$, from the start of 2026 | Forster et al.; Global Carbon Budget 1,055 |
| Peak | GHG emissions peak between 2020 and "at the latest before 2025" in 1.5 °C pathways | IPCC AR6 Synthesis Report B.6.1 |

The 43% and 60% figures are the numbers the first global stocktake reaffirmed in 2023.

## 2. How fast would emissions have to fall?

The pathway figures are reductions from 2019. Observed emissions have since risen (fossil $CO_2$ was 4.1% above 2019 in 2024), so the required rate from today is steeper than the headline percentage suggests. The table converts each $CO_2$ pathway into the **constant annual percentage cut from the 2024 level** that would reach it:

| Pathway ($CO_2$, from 2019) | By 2030 | By 2035 | By 2040 |
|---|---|---|---|
| **1.5 °C** median reduction | –48% | –65% | –80% |

| Pathway ($CO_2$, from 2019) | By 2030 | By 2035 | By 2040 |
|---|---|---|---|
| Constant cut needed each year from 2024 | **10.9% a year** | 9.4% a year | 9.8% a year |
| **2 °C** median reduction | –22% | –37% | –51% |
| Constant cut needed each year from 2024 | **4.7% a year** | 4.5% a year | 4.6% a year |

(Calculated on fossil $CO_2$ including bunkers, 38.6 $GtCO_2$ in 2024 and 37.1 in 2019. Including land use changes the rates by under 0.3 percentage points.)

For scale:

- **The largest annual fall since 1960 was in 2020, the pandemic year: –5.2%** in the platform's fossil $CO_2$ series (the IPCC gives –5.8% for $CO_2$ from fossil fuels and industry). The next largest falls were about –3.0% (1992) and –2.7% (1981).
- A **sustained** 4.5–4.7% a year, the 2 °C requirement, is roughly the 2020 pandemic fall repeated every year for 15 years. The 1.5 °C requirement of about 10% a year is roughly twice that.
- These are **median pathway** requirements. The ranges in the IPCC's table are wide (for the 1.5 °C 2030 $CO_2$ reduction, –36% to –69%).

## 3. The platform's pathways against the targets

The platform's scenarios ($CO_2$, 2025–2040) are shown in Pathways. In summary, by 2035 relative to 2019:

| | BAU | Moderate (–2%/yr) | Aggressive (–5%/yr) | IPCC 2 °C median | IPCC 1.5 °C median |
|---|---|---|---|---|---|
| $CO_2$ change by 2035 | +16% | –5% | –31% | –37% | –65% |

No platform scenario meets even the 2 °C median pathway. The platform's scenarios are illustrative bounds, not forecasts of policy, but the arithmetic is informative. Because the cuts are applied to a business-as-usual path that is still growing, the Aggressive scenario's 5% cut is a net fall of only about 3.8% a year between 2024 and 2040 (38.6 to 20.8 $GtCO_2$). Sustained for 16 years, that is faster than every single year in the past 60 except the 2020 pandemic, and it is still short of the roughly 4.6% a year the 2 °C median requires.

## 4. When would the carbon budgets run out?

Cumulative $CO_2$ from the platform's scenarios (fossil and cement, plus land use held flat at about 4.7 $GtCO_2$ a year) can be run against the remaining budgets. Each cell is the first year the scenario's cumulative emissions since the start of 2026 reach the budget, at a 50% likelihood.

| Budget from the start of 2026 (Forster et al. / Global Carbon Budget) | BAU | Moderate | Aggressive |
|---|---|---|---|
| 1.5 °C: 130 / 170 $GtCO_2$ | 2028 / 2029 | 2029 / 2029 | 2029 / 2030 |
| 1.7 °C: 500 / 525 $GtCO_2$ | 2036 / 2037 | 2038 / 2038 | not by 2040 (496 emitted by 2040) |

| Budget from the start of 2026 (Forster et al. / Global Carbon Budget) | BAU | Moderate | Aggressive |
|---|---|---|---|
| 2 °C: 1,050 / 1,055 $GtCO_2$ | not by 2040 | not by 2040 | not by 2040 |

(Each year pair reads the result under Forster et al.'s budget / the Global Carbon Budget's.) Cumulative emissions from 2026 to 2040 are 702 $GtCO_2$ for BAU, 608 for Moderate and 496 for Aggressive.

How to read this:

- **The 1.5 °C budget is used up around 2028–2030 in every scenario**, whichever of the two budget estimates is used. The difference between the 130 and 170 figures moves the date by at most a year, and the scenarios themselves differ by at most about two years (2028 for BAU, 2030 for Aggressive): the budget is consumed faster than any of these pathways can bend.
- **The 1.7 °C budget is passed in the late 2030s except under the Aggressive scenario**, which reaches it only at the end of 2040.
- **The 2 °C budget lasts beyond 2040 in all three.**
- **Using up the budget is not the same as reaching 1.5 °C.** The published budgets include uncertainty in how warming responds to $CO_2$ only, assume a further 0.10 °C of future non-$CO_2$ warming, and are for a 50% chance. The overshoot the COP30 decision acknowledged is related: the budget is spent before emissions reach zero.

## 5. What the targets imply for the platform's own temperature path

The platform's BAU temperature path reaches a level of 1.5 °C in 2033 and Aggressive in 2034 (see Pathways §5). That is consistent with the IPCC's finding that the first 20-year average to reach 1.5 °C is centred in the first half of the 2030s in all but its highest-emissions scenario, and with Forster et al.'s estimate of around 2030 for human-induced warming. The Paris goal concerns the long-term level, so a single warm year above 1.5 °C (2024 already was) does not by itself mean the goal is breached.

## See also

- The Paris Agreement and the COP Process
- Country NDCs
- Scenarios and Temperature Mapping